

4.1 Simultaneous Equations:

⇒ When 2 or more variables must solve each seperate equation.

① Elimination → Make the equations have the same coefficient of a variable (can be negative) & add / subtract the equations

② Substitution → Express one variable in terms of another, and sub into the other equation present.

For linear & non-linear sim equations, only substitution can be used.

4.2 Inequalities:

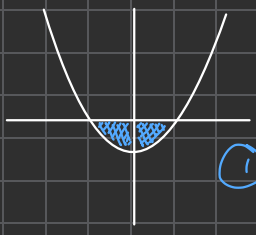
Used to find a range of possible solutions
Can be represented graphically or on a number line.

Same method as solving equations, just be careful of signs & what they actually represent graphically (easier than algebra)
Linear inequalities can be solved regularly.
For quadratics, think of where the solution fits graphically.
Set notation:

2 regions: $\{x : x < a\} \cup \{x : x > b\}$
1 region: $\{x : x > a\} \cap \{x : x < b\}$

Inequalities can be shaded graphically to show where solutions lie.

$x^2 - 1 \leq 0$
$(x-1)(x+1) \leq 0$



① $\{x : x \geq -1\} \cap \{x : x \leq 1\}$

② if $x < 0$, smallest $= -1$
if $x > 0$, smallest $= 1$

$-1 \leq x \leq 1$